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(54) **CLEANING SYSTEM WITH CROP DIVIDER
STROKER ARRAY**

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A01F 12/46 (2006.01)
A01F 12/44 (2006.01)

Primary Examiner — Joseph M Rocca
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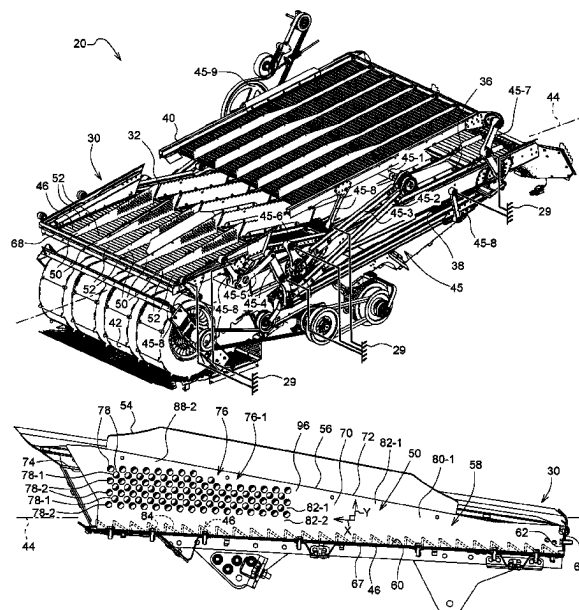
(52) **U.S. Cl.**
CPC **A01F 12/46** (2013.01); **A01F 12/44**
(2013.01)

(57) **ABSTRACT**

(58) **Field of Classification Search**
CPC A01F 12/44; A01F 12/46; A01F 12/32;
A01F 12/30
USPC 460/101
See application file for complete search history.

A cleaning system for an agricultural combine harvester includes a pan and a crop divider. The pan includes a stepped floor to receive crop material thereon and is arranged for fore-aft reciprocating movement to advance the crop material in a crop-processing direction. The crop divider is mounted with the pan for fore-aft reciprocating movement therewith and extends in a fore-aft manner and above the

(Continued)



stepped floor to partition the crop material laterally. The crop divider includes a body and a periphery disposed about the body. The body includes a multi-dimensional, clustered array of crop strokers extending laterally to engage the crop material to advance the crop material in the crop-processing direction during fore-aft reciprocating movement of the pan.

17 Claims, 12 Drawing Sheets

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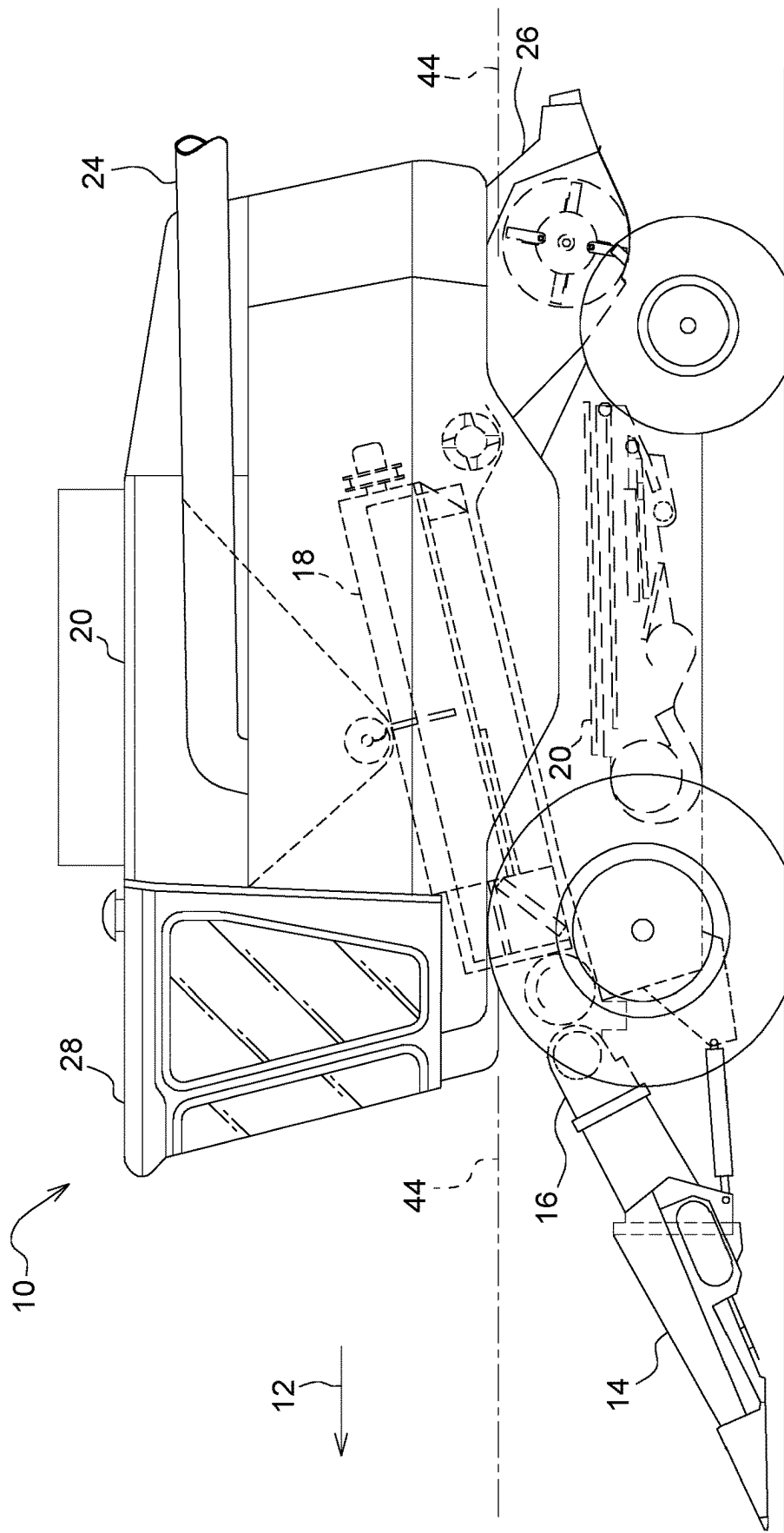


FIG. 1

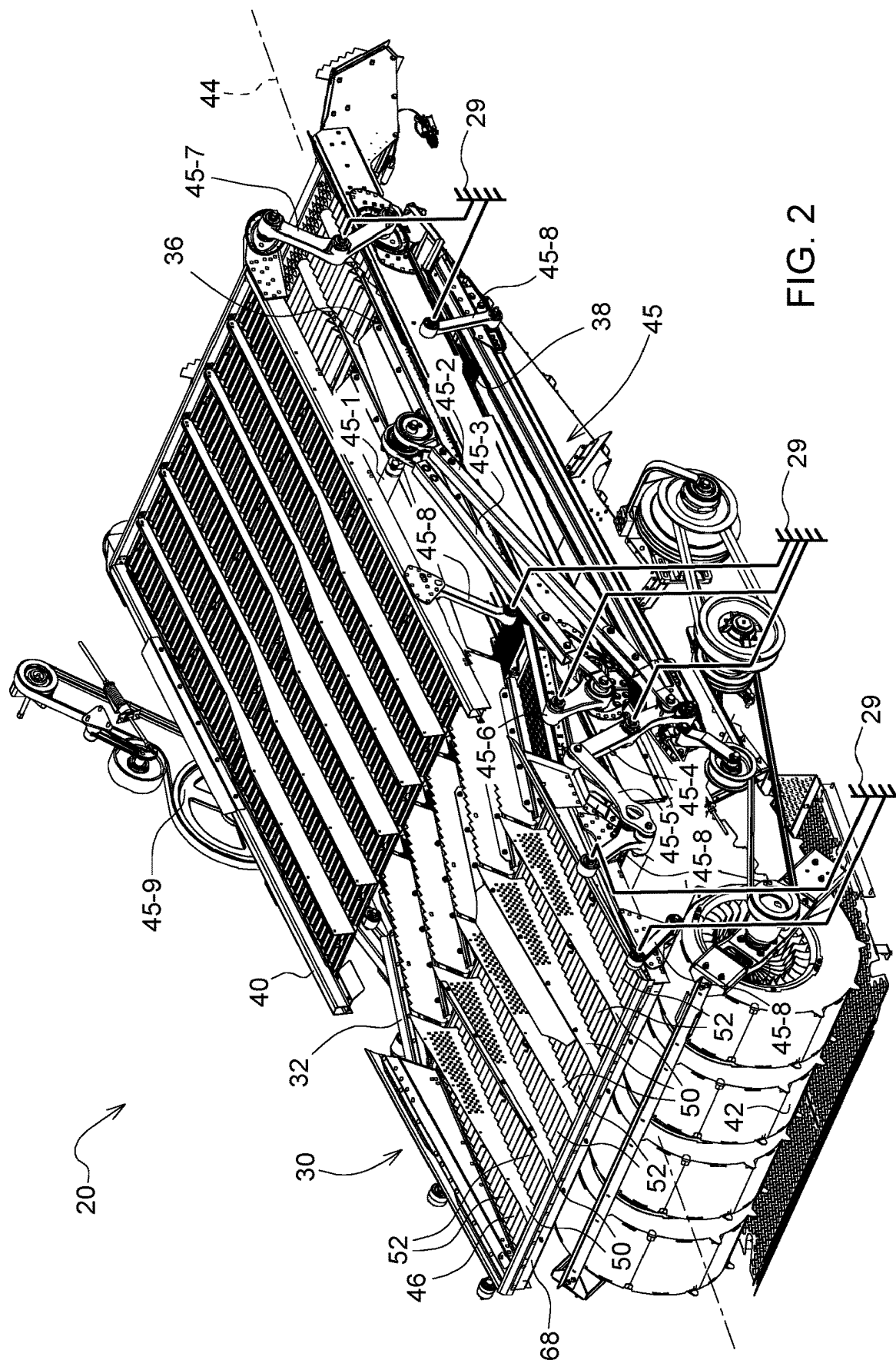


FIG. 2

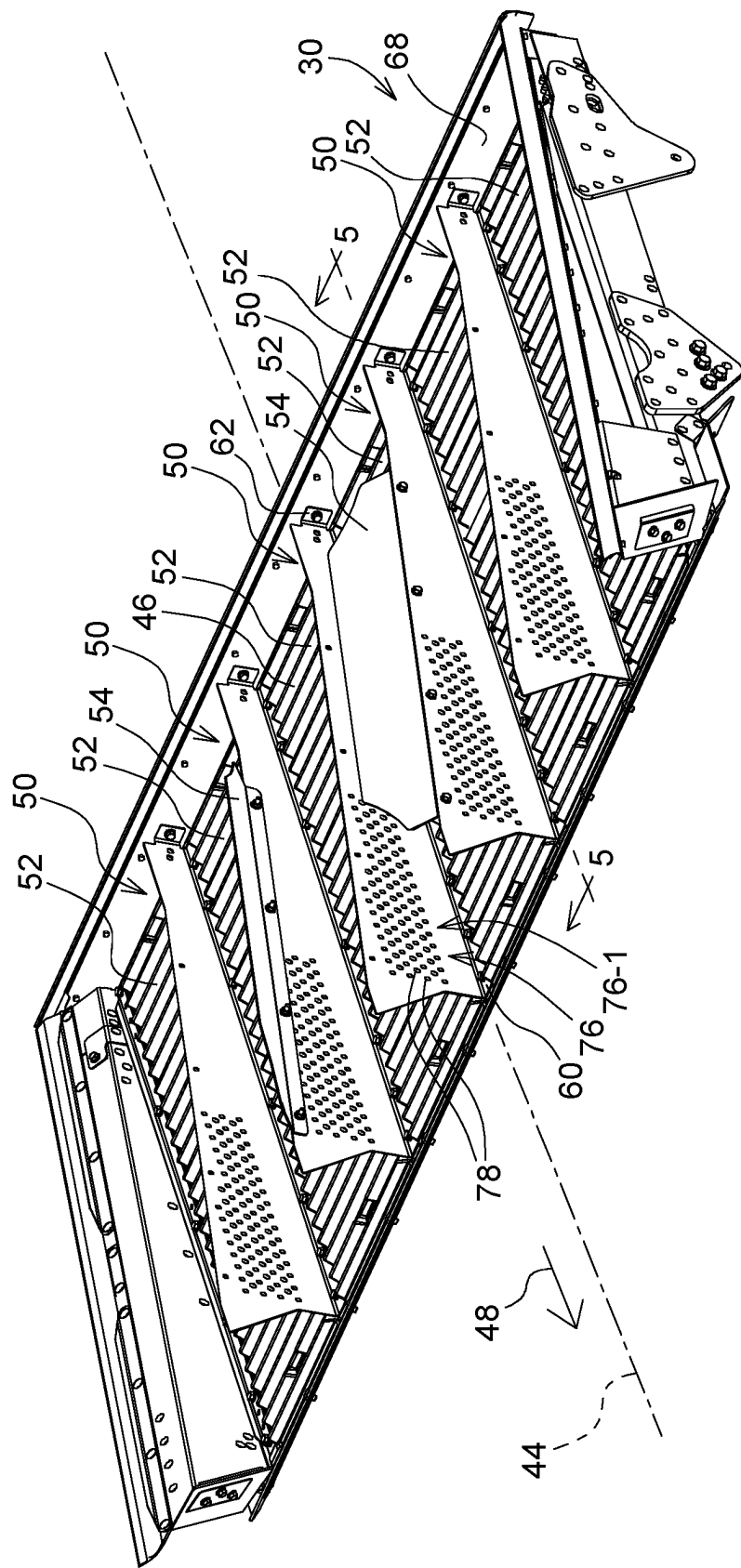


FIG. 3

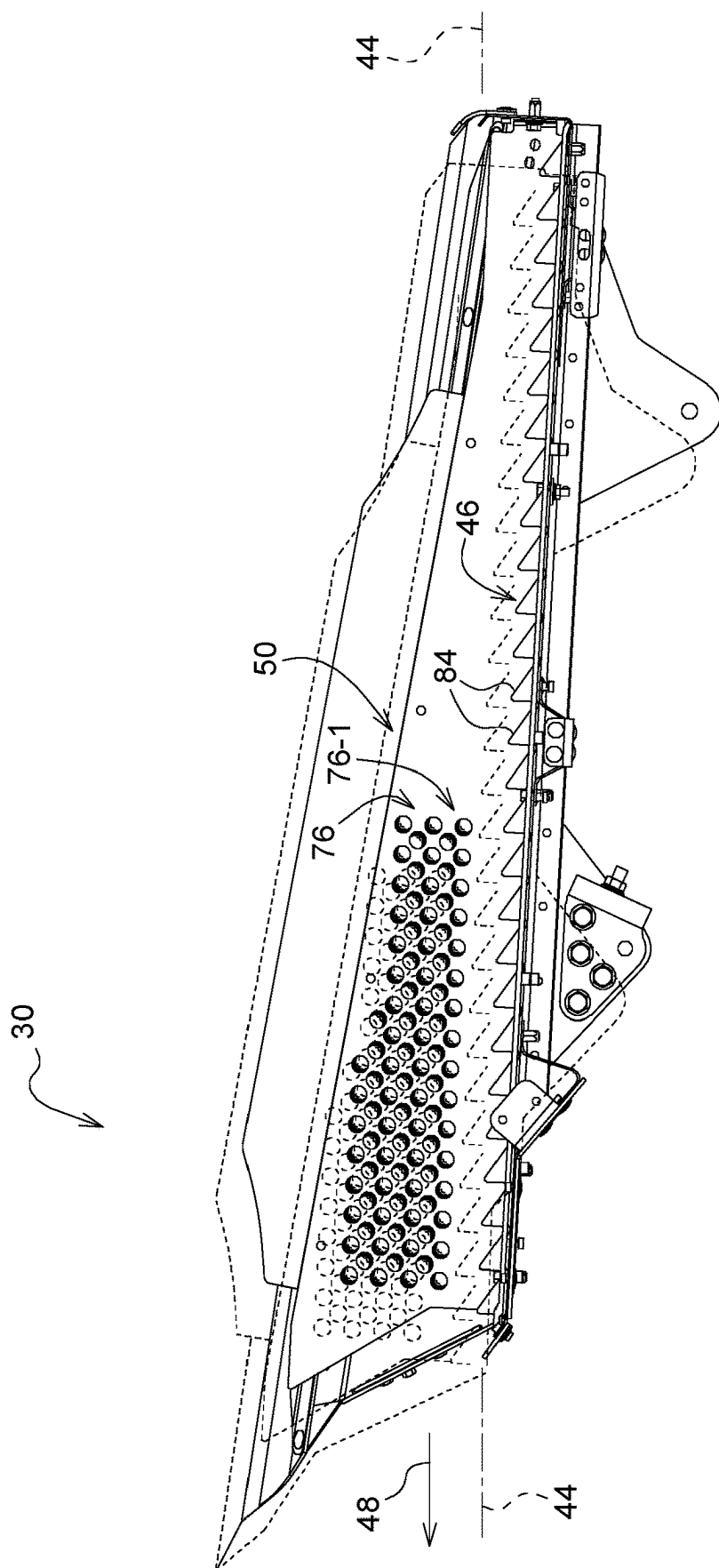


FIG. 3A

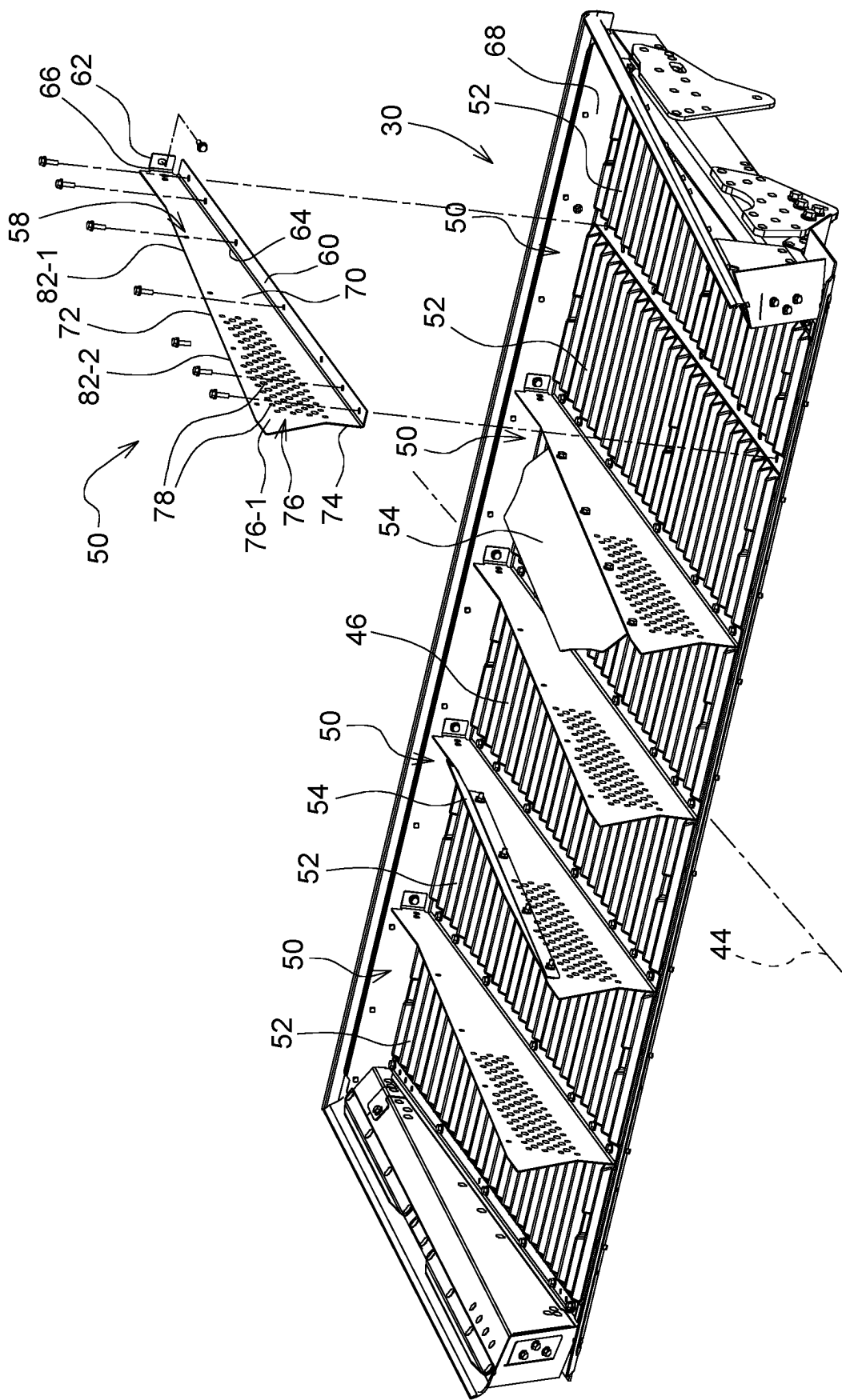


FIG. 4

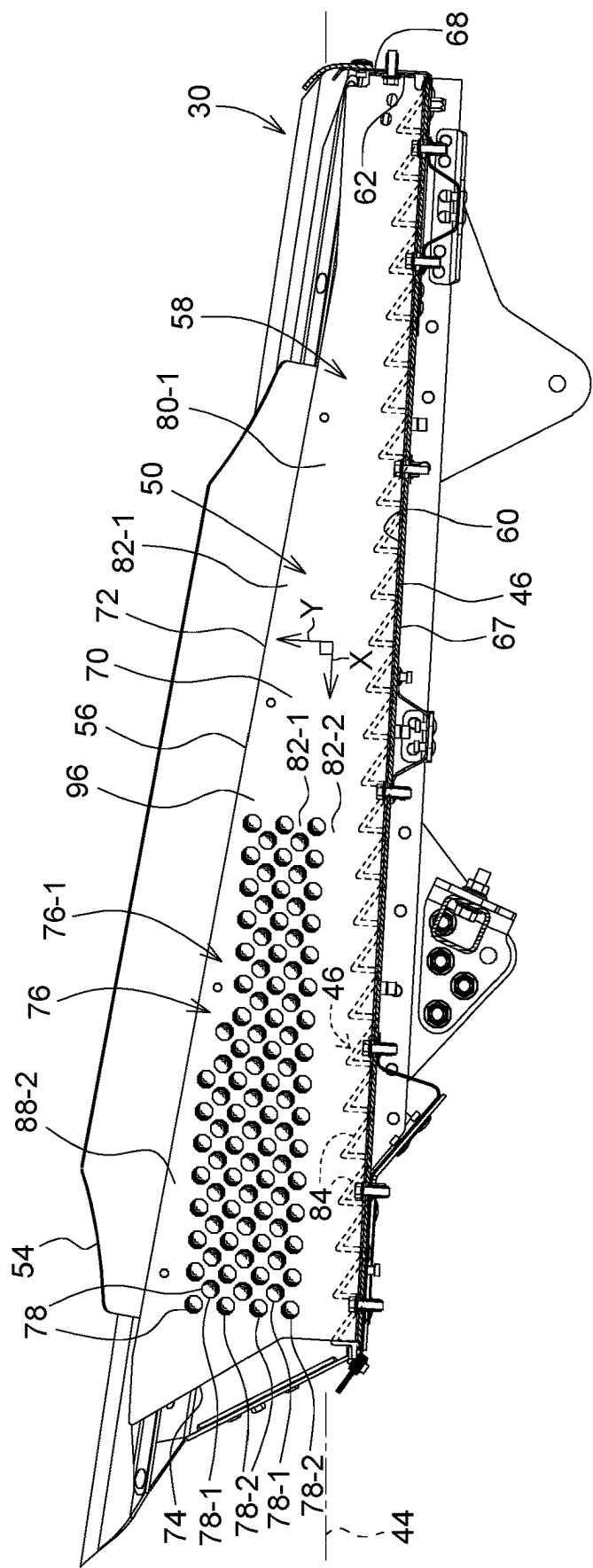


FIG. 5

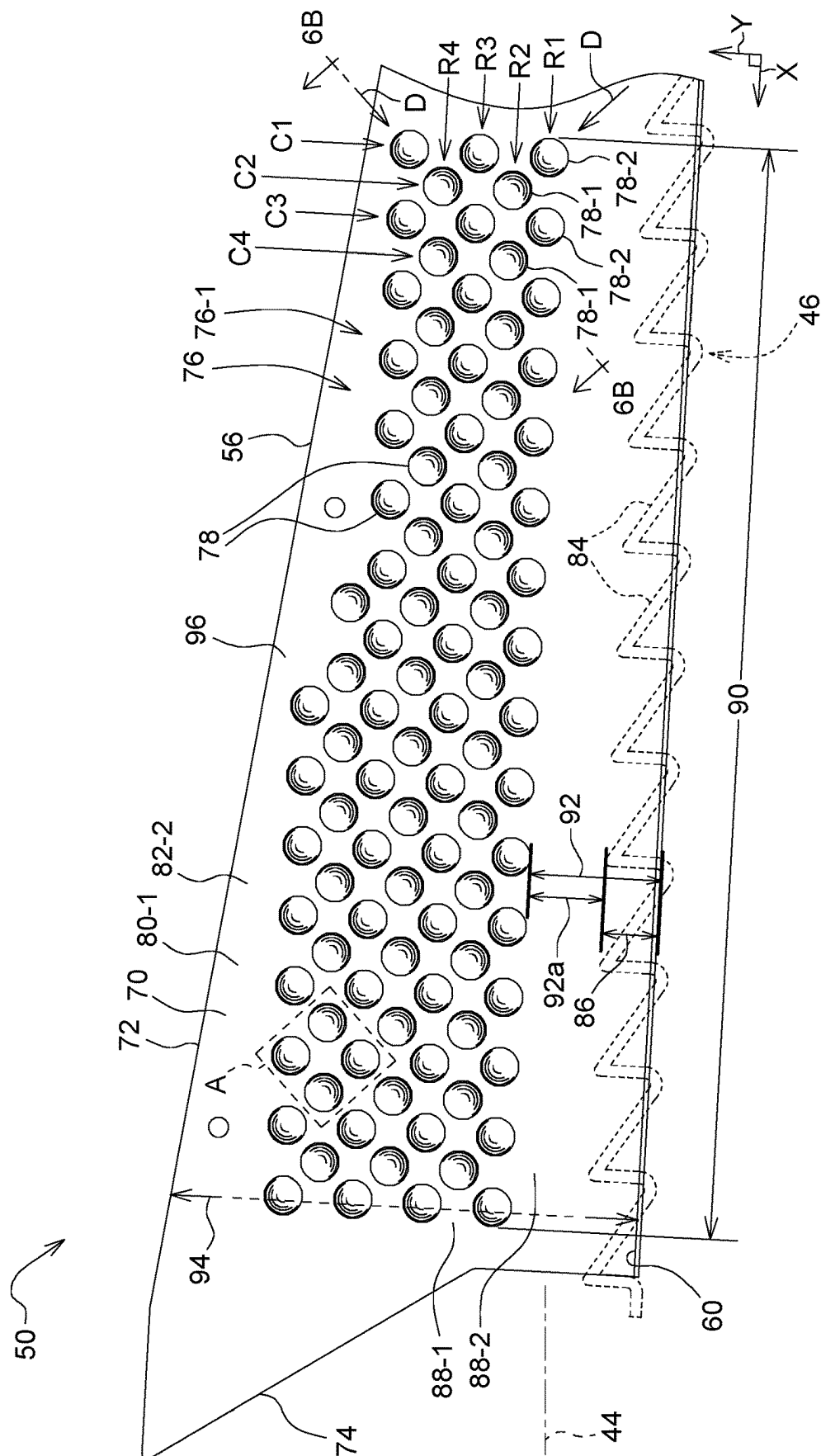


FIG. 6

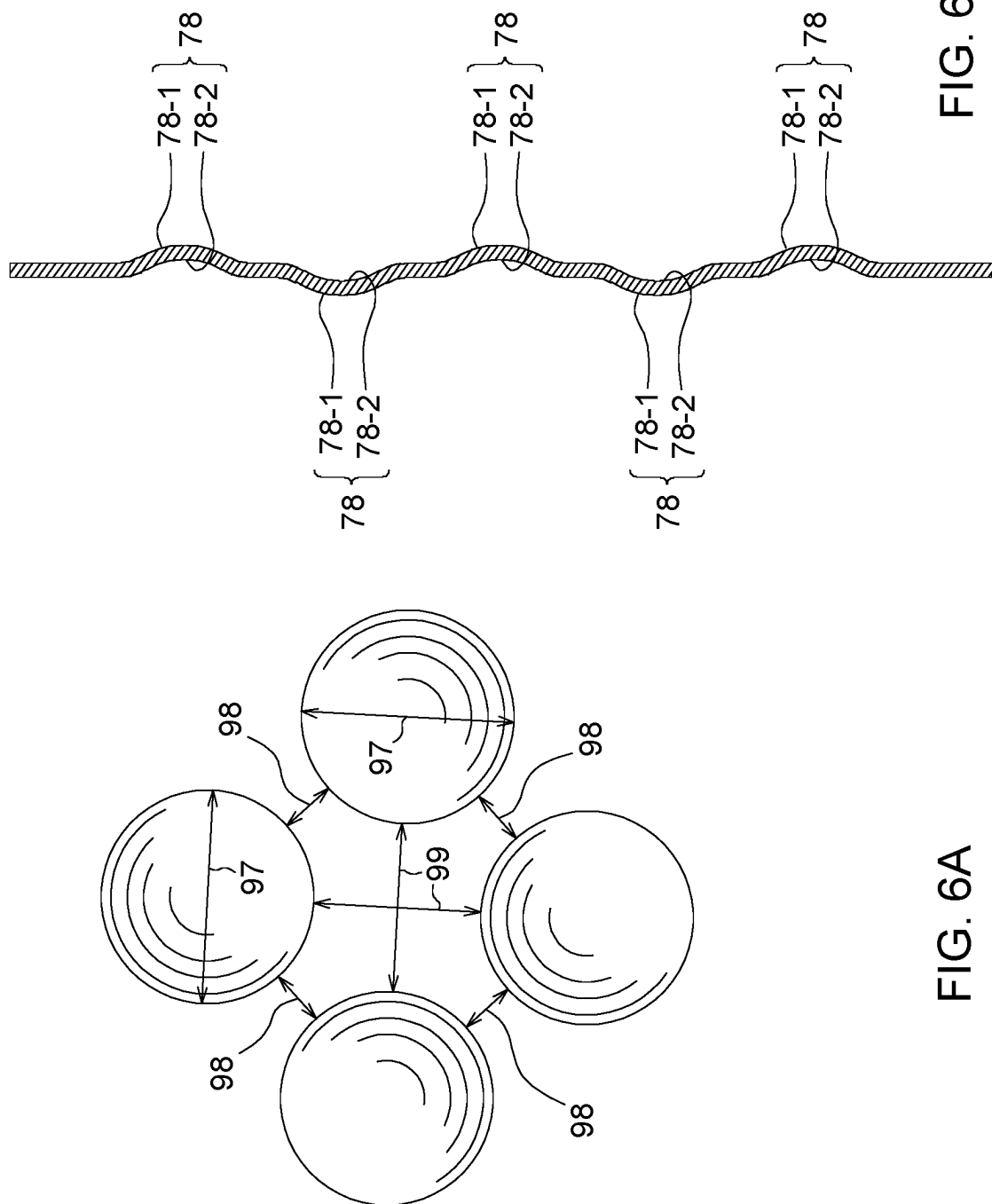


FIG. 6A

FIG. 6B

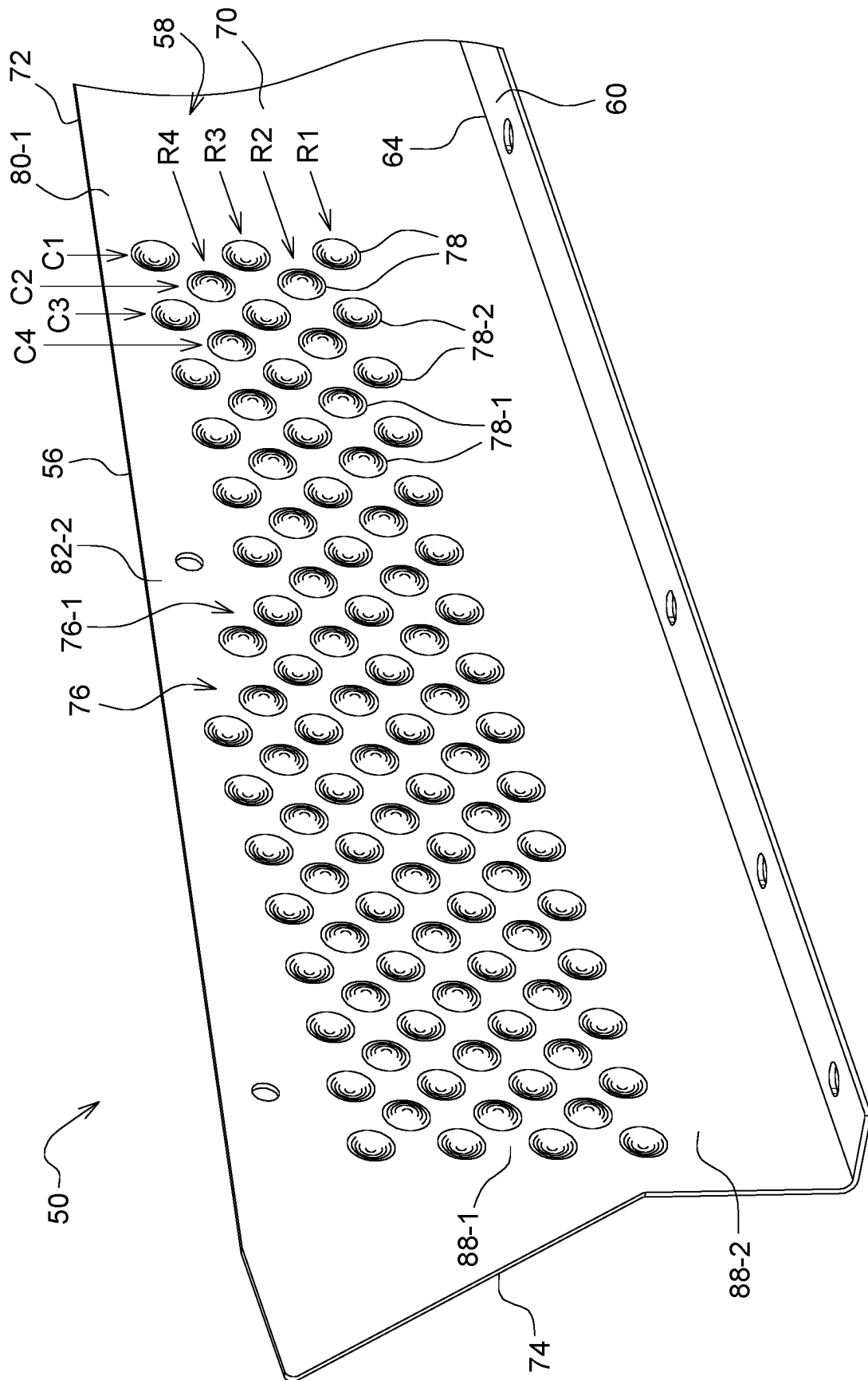


FIG. 7

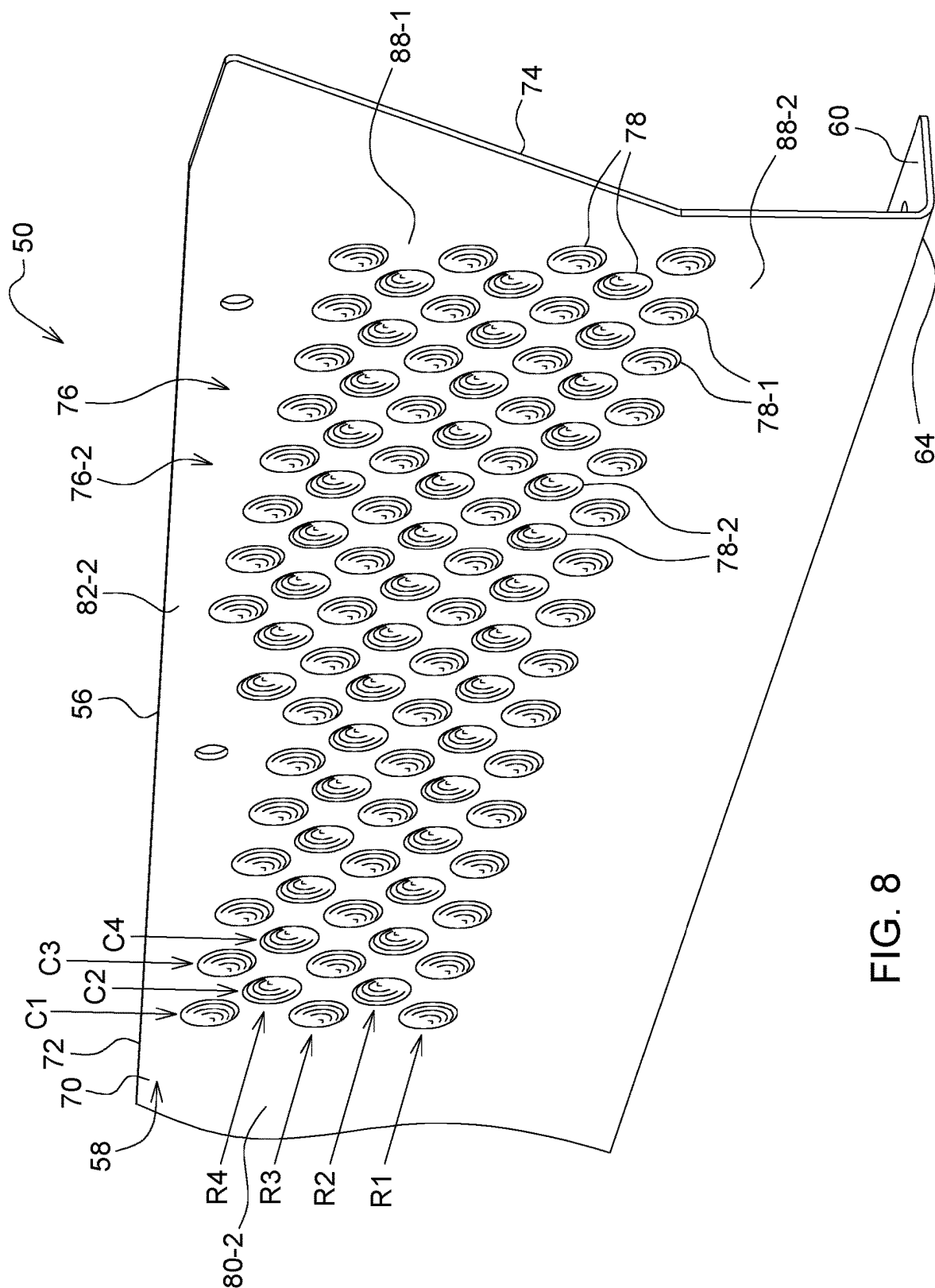


FIG. 8

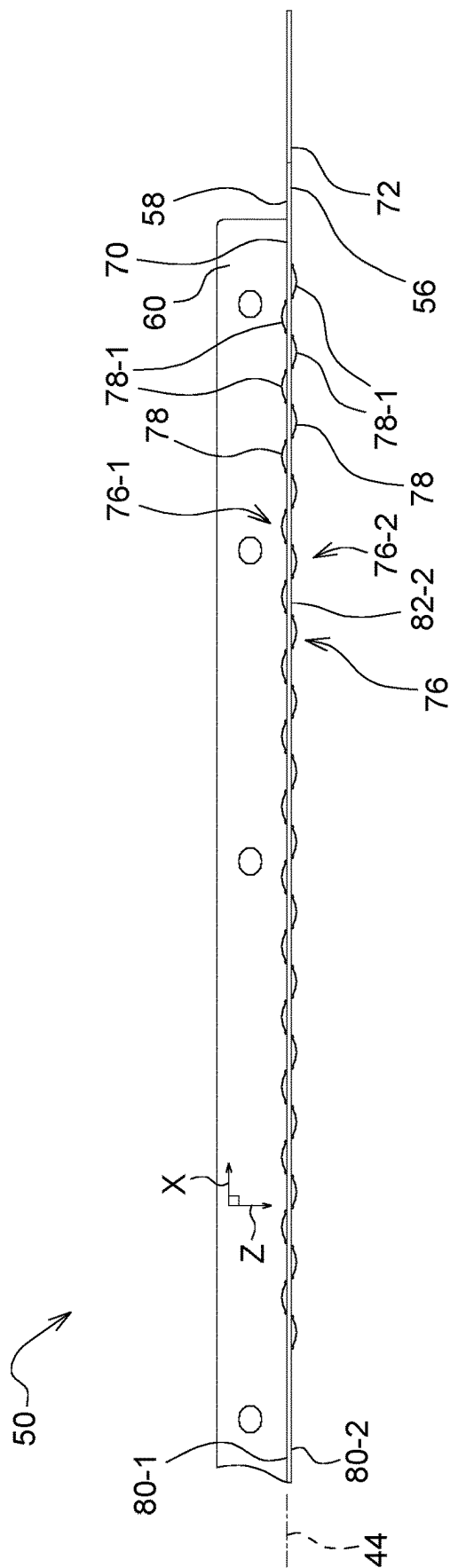
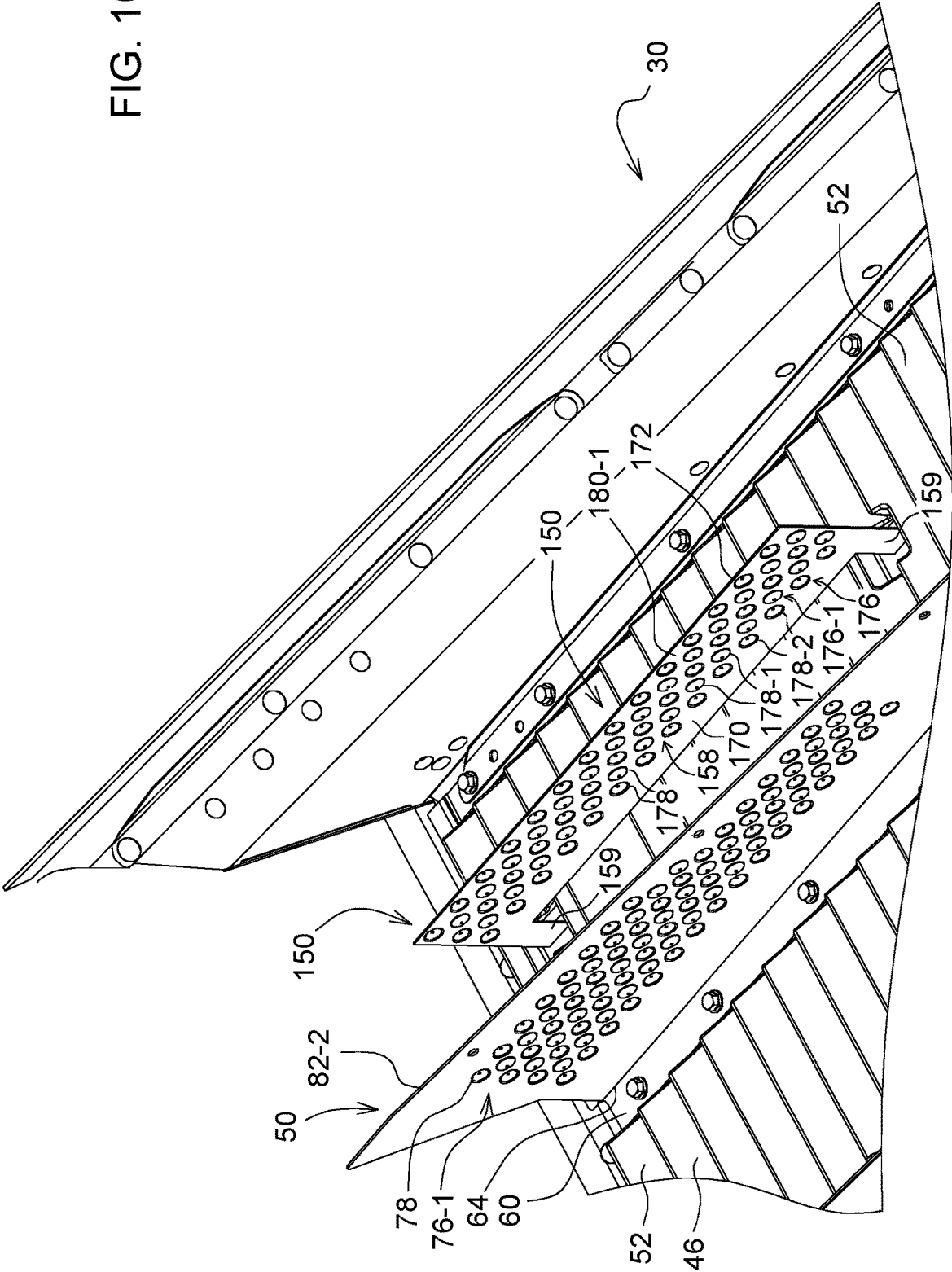


FIG. 9

FIG. 10



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CLEANING SYSTEM WITH CROP DIVIDER STROKER ARRAY

TECHNICAL FIELD

The present disclosure relates to a cleaning system for an agricultural combine harvester.

BACKGROUND

An agricultural combine harvester intakes crop gathered from a field and processes the crop to separate grain from material other than grain (MOG). Combine harvesters have cleaning systems to separate grain from smaller pieces of MOG. Cleaning systems sometimes have crop dividers to partition crop material laterally into bays.

SUMMARY

According to an aspect of the present disclosure, there is disclosed a cleaning system for an agricultural combine harvester. The cleaning system includes a pan and a crop divider. The pan includes a stepped floor to receive crop material thereon and is arranged for fore-aft reciprocating movement to advance the crop material in a crop-processing direction. The crop divider is mounted with the pan for fore-aft reciprocating movement therewith and extends in a fore-aft manner and above the stepped floor to partition the crop material laterally. The crop divider includes a body and a periphery disposed about the body. The body includes a multi-dimensional, clustered array of crop stokers extending laterally to engage the crop material to advance the crop material in the crop-processing direction during fore-aft reciprocating movement of the pan.

The above and other features will become apparent from the following description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The detailed description of the drawings refers to the accompanying figures in which:

FIG. 1 a diagrammatic side elevational view, with portions broken away, showing an agricultural combine harvester with various functional systems including a cleaning system underlying a threshing and separating system;

FIG. 2 is a perspective view showing the cleaning system, the cleaning system including a front step pan (on the left) and a number of crop dividers partitioning the front step pan into bays with crop material;

FIG. 3 is an enlarged perspective view showing the front step pan and the crop dividers mounted with the front step pan;

FIG. 3A is a side elevational view showing the front step pan and the crop dividers in a forwardmost position (in solid) and a rearwardmost position (in phantom);

FIG. 4 is an exploded perspective view showing the front step pan and crop dividers;

FIG. 5 is a sectional view taken along lines 5-5 of FIG. 3;

FIG. 6, is an enlarged side elevational view, with portions broken away, showing a crop divider with an array of crop stokers configured as bumps;

FIG. 6A is an enlarged side elevational view of region A of FIG. 6;

FIG. 6B is a sectional view taken along lines 6B-6B of FIG. 6;

FIG. 7 is a perspective view, with portions broken away, showing a lateral side of the crop divider;

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FIG. 8 is a perspective view, with portions broken away, showing an opposite lateral side of the crop divider;

FIG. 9 is a top plan view of the crop divider showing crop stokers projecting laterally from either lateral side of the crop divider; and

FIG. 10 is a perspective view showing an intermediate crop divider with an array of crop stokers.

DETAILED DESCRIPTION

Referring to FIG. 1, an agricultural combine harvester 10 is configured to move in a forward or harvest direction of travel 12 over a field to harvest crop material from the field. The harvester 10 processes the crop, separating grain from residual crop material known as material other than grain or MOG (e.g., straw, stalks, cobs, leaves, chaff).

In general, the harvester 10 may include front-end equipment 14 to cut, gather, and transport crop rearwardly (some front-end equipment 14 may not cut crop, as in the case of a belt pick-up unit), a feederhouse 16 to advance crop material received from the equipment 14 into the body of the harvester 10, a threshing and separating system 18 to thresh crop material and separate grain from material other than grain (MOG), a cleaning system 20 (also known as a “cleaning shoe”) to separate grain from chaff and other MOG, a clean grain elevator (not shown) to elevate clean grain to a storage bin 22, an unloader 24 to unload clean grain from the storage bin 22 to another location, and a residue system 26 to process and distribute crop residue back onto the field. A person can control the harvester 10 from an operator's station 28 of the harvester 10. The harvester 10 may be configured in a wide variety of ways.

The threshing and separating system 18 includes an axial rotor and a concave assembly. The rotor and the concave assembly cooperate to provide the threshing and separating system with a front threshing zone for threshing grain and a rear separating zone for separating threshed grain from MOG. In some examples, the threshing and separating system 18 includes a second rotor (not shown) and a second concave assembly (not shown), which cooperate to provide the threshing and separating system 18 with a second front threshing zone and a second rear separating zone. In such a case, the rotors are similar to one another in structure and function, and the concave assemblies are similar to one another in structure and function. The threshing and separating system 18 may be configured in a wide variety of ways.

Referring to FIG. 2, the cleaning system 20 includes a front step pan 30 (which may also be referred to as a preparation pan), an inclined front chaffer 32 (which may also be referred to as a sieve), a lower step pan (not shown), a chaffer 36 (which may also be referred to as a sieve), a sieve 38, a return pan 40, and a fan assembly 42 (e.g., including four fans). The lower step pan is mounted with the chaffer 36 so as to be positioned in front of the chaffer 36 and below the front chaffer 32. The front step pan 30, the front chaffer 32, the lower step pan, the chaffer 36, the sieve 38, and the return pan 40 are arranged for fore-aft reciprocating movement in a fore-aft dimension 44 of the combine harvester 10 to process and advance crop material.

A drive mechanism 45 imparts the fore-aft reciprocating movement. The drive mechanism 45 includes a rotary drive 45-1 and a linkage including a first drive link 45-2, a second drive link 45-3, a front rocker link 45-4, a pan link 45-5, a chaffer link 45-6, a rear rocker link 45-7, and a number of support links 45-8. The linkage is positioned on the left side of the combine harvester 10. The rotary drive 45-1 is rotated

by a belt-driven pulley 45-9 positioned on the right side of the combine harvester 10. The first and second drive links 45-2, 45-3 are coupled eccentrically to the rotary drive 45-1 such that rotation of the rotary drive 45-1 imparts a reciprocating motion to the first and second drive links 45-2, 45-3. The first drive link 45-2 is coupled to the front rocker link 45-4 which is coupled at a first end to the sieve 38 to reciprocate the sieve 38 and at an opposite second end to the pan link 45-5 that reciprocates the front step pan 30 and the front chaffer 32. As such, the sieve 38 reciprocates in a manner opposite to the pan 30 and the front chaffer 32. The second drive link 45-3 is coupled to the chaffer link 45-6 which is coupled to the chaffer 36 to reciprocate the chaffer 36. The rear rocker link 45-7 is coupled at a first end to the chaffer 36 and at an opposite second end to the return pan 40 to reciprocate the return pan 40 in a manner opposite to the chaffer 36 in response to reciprocation of the chaffer 36. The support links 45-8 support the front step pan 30, the front chaffer 32, the lower step pan, the chaffer 36, the sieve 38, and the return pan 40 on the support structure 29 of the combine harvester 10. A similar linkage is positioned on the right side of the combine harvester.

The front step pan 30, the front chaffer 32, the lower step pan, and the chaffer 36 reciprocate in phase as a first unit. The sieve 38 and the return pan 40 reciprocate in phase as a second unit. The first and second units reciprocate 180 degrees out of phase with one another. The fore-aft reciprocating movement of the first and second units is relative to the support structure 29. It is to be appreciated that the front step pan 30, the front chaffer 32, the lower step pan, the chaffer 36, the sieve 38, and the return pan 40 may be driven for reciprocating movement in any suitable manner.

The front step pan 30 is positioned under a front portion of the threshing and separating system 18 to receive crop material on a stepped floor 46 of the pan 30. The pan 30 is configured for fore-aft reciprocating movement in the fore-aft dimension 44 to advance crop material in a crop-processing direction 46 toward the front chaffer 32. The crop-processing direction 46 is rearward with respect to the front step pan 30.

The return pan 40 is positioned under a rear portion the threshing and separating system 18 to receive crop material on a stepped floor of the return pan 40. Tailings may also be routed back to the return pan 40 for further processing by the cleaning system 20.

The pan 30 is configured for fore-aft reciprocating movement in the fore-aft dimension 44 to advance crop mat in a crop-processing direction toward the front chaffer 32. The crop-processing direction is forward with respect to the return pan 40.

Crop material may pass rearwardly from the front chaffer 32 to the chaffer 36. Grain and MOG may pass through the front chaffer 32 to the lower pan 34 and the chaffer 36. Grain and smaller pieces of MOG received from the front chaffer 32 and lower pan 34 may pass through the chaffer 32 to the sieve 38, which further filters for grain. The fan assembly 42 blows air rearwardly through and across the chaffers 32, 36 and sieve 38 to advance MOG to the residue system 26 for discharge from the combine harvester 10.

Referring to FIGS. 3-5, the cleaning system 20 includes one or more crop dividers 50. For example, there are five crop dividers 50. It is to be appreciated that there could be any suitable number of crop dividers 50.

The crop dividers 50 are mounted with the pan 30 for fore-aft reciprocating movement therewith. In such a case, the pan 30 and the crop dividers 50 mounted therewith have a forwardmost position and a rearwardmost position, which

is rearward and may be somewhat upward from the forwardmost position (FIG. 3A). The crop dividers 50 partition the pan 30 into crop-receiving bays 52 (e.g., six bays).

Each crop divider 50 extends in a fore-aft manner in the fore-aft dimension 44 and above the stepped floor 46 of the pan 30 to partition the crop material laterally to inhibit lateral movement of crop material to one lateral side of the pan 30 or the other to foster lateral distribution of crop material in the cleaning system for cleaning effectiveness. The crop divider 50 partitions the pan 30 into laterally adjacent bays 52. A divider extension 54 may be mounted (e.g., bolted) to the crop divider 50 along a top edge 56 of the crop divider 50, as shown, for example, with respect to two crop dividers 50.

The crop divider 50 includes a panel 58, an attachment flange 60, and an attachment tab 62. The attachment flange 60 is coupled to and extends along a bottom edge 64 of the panel 58. The attachment flange 60 is coupled to the floor 46 and an underlying sub-structure 67 of the pan 30 with fasteners. Such fasteners may include, for example, hex flange bolts (e.g., six hex flange bolts) threaded to corresponding rivet nuts (e.g., six rivet nuts) and a hex flange bolt (e.g., third hex flange bolt from front of pan 30) received by a washer and threaded to a nut. The attachment tab 62 is coupled to a front edge 66 of the panel 58. The attachment tab 62 is coupled to a front wall 68 of the pan 30 with a fastener. Such fastener may include, for example, a hex flange bolt threaded to corresponding rivet nut. For ease of illustration, threads are not shown, and the rivet nuts are shown in their uncompressed state. The presence of threads and rivet nut compression for locking in place are to be understood. The crop divider 50 is configured, for example, as a formed piece of sheet metal (e.g., stamped sheet metal), although it may be configured in a wide variety of ways.

Referring to FIGS. 5-9, the panel 58 includes a body 70 and a periphery 72 disposed about the body 70. The periphery 72 includes the top edge 56, the bottom edge 64, the front edge 66, and a rear edge 74. The edges 56, 64, 66, 74 cooperate to provide the periphery 72.

The body 70 includes a multi-dimensional, clustered array 76 of crop stalkers 78 extending laterally to engage crop material to advance the crop material in the crop-processing direction 48 (e.g., rearward) during fore-aft reciprocating movement of the pan 30. The array 76 includes a first array 76-1 of crop stalkers on a first lateral side 80-1 of the body 70 and a second array 76-2 of crop stalkers on an opposite second lateral side 80-2 of the body 70.

Each array 76-1, 76-2 is itself a multi-dimensional, clustered array of crop stalkers. The first array 76-1 is positioned on the first lateral side 80-1 and extends laterally to engage crop material in a first bay 52 of the bays 52 to advance the crop material present in that bay 52 in the crop-processing direction 48 during fore-aft reciprocating movement of the pan 30. The second array 76-2 is positioned on the second lateral side 80-2 and extends laterally to engage crop material in a second bay 52 of the bays 52 to advance the crop material present in that bay 52 in the crop-processing direction 48 during fore-aft reciprocating movement of the pan 30.

The crop stalkers 78 of the array 76 are clustered together to promote the crop-advancing effectiveness of the array 76. The crop divider 50 includes an upstream half 82-1 and a downstream half 82-2 downstream of the upstream half 82-1 in the crop-processing direction 48. The crop stalkers 78 are clustered, for example, in the downstream half 82-1.

The crop divider 50 has an X dimension, a Y dimension, and a Z dimension. The X dimension is parallel to the floor

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46 in the direction of the length of the crop divider 50. The Y dimension is orthogonal to the floor 46 in an upward direction. The Z dimension is orthogonal to the X and Y dimensions in a direction of the thickness of the crop divider 50. The floor 46 has steps 84 with a step height 8 in the Y dimension.

The body 70 of the crop divider 50 includes an upper zone 88-1 and a lower zone 88-2. The upper zone 88-1 is textured with the array 76 of crop stalkers 78 and is defined by the extent of the array 76. The lower zone 88-2 is positioned between the upper zone 88-1, more particularly the array 76, and the stepped floor 46, and is devoid of the array 76. The upper zone 88-1 is rougher than the lower zone 88-2 to promote engagement between the crop divider 50 and the crop material to advance the crop material during the crop-advancement stroke (rearward stroke) of the pan 30, which stroke is generally rearward and upward. The lower zone 88-2 presents less friction than the upper zone 88-1 so as to reduce engagement between the crop divider 50 and the crop material during the return stroke (forward stroke) of the pan 30, which stroke is generally forward and downward. The lower zone 88-2 is defined in the X dimension by a length 90 of the array 76 and in the Y dimension by a distance 92 between the attachment flange 60 and the array 76. The distance 92 is at least two times the step height 86. A portion of the distance 92 is a distance 92a defined in the Y dimension between the steps 84 of the stepped floor 46 and the array 76. The distance 92a is greater than or equal to the value of the step height 86.

The array 76 increases in height 94 in the Y dimension from the stepped floor 46 as the array 76 extends in the crop-processing direction 48, to further promote engagement between the crop divider 50 and the crop material to advance crop material during the crop-advancement stroke of the pan 30. The crop divider 50 includes an expansion zone 96 in which the top edge 56 increases in height 94 in the Y dimension from the stepped floor 46 as the crop divider 50 extends in the crop-processing direction 48. The expansion zone 94 includes at least most of the array 76.

The array 76 is arranged multi-dimensionally. For example, the crop stalkers 78 are arranged relative to one another in at least two orthogonal reference dimensions, e.g., dimensions X and Y. The array 76 includes rows (R1, R2, . . . Rn) of crop stalkers 78 and columns (C1, C2, . . . Cn) of crop stalkers 78. The rows may be arranged in the X dimension, and the columns may be arranged in the Y dimension. The crop stalkers 78 of each row are positioned at regular intervals (e.g., equal intervals), and the crop stalkers 78 of each column are positioned at regular intervals (e.g., equal intervals).

The crop stalkers 78 are clustered in proximity to one another to maximize the rough surface area. Immediately adjacent rows of crop stalkers 78 are alternately staggered relative to one another, and immediately adjacent columns of crop stalkers 78 are alternately staggered relative to one another. Such staggering accommodates more crop stalkers 78 in a given surface area, thereby maximizing the rough surface area. In other embodiments, immediately adjacent rows may not be staggered, and immediately adjacent columns may not be staggered.

Each crop stalker 78 has a maximum size 97. In the illustrated embodiment, the crop stalker 78 is round, in which case, the maximum size 97 is the diameter of the crop stalker 78. Immediately adjacent crop stalkers 78 are spaced apart from one another by an offset that is less than the maximum size 97. For example, immediately adjacent crop stalkers 78 of each diagonal D of crop stalkers 78 are spaced

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apart from one another by a first offset 98 that is less than the maximum size 97. Immediately adjacent crop stalkers 78 of each row are spaced apart from one another by a second offset 99, and immediately adjacent crop stalkers 78 of each column are spaced apart from one another by the second offset 99. The second offset 99 is less than the maximum size 97, although the second offset 99 may be larger than the first offset 98 due to the staggering of rows and columns. It is to be appreciated that the crop stalkers 78 may be non-round.

Each crop stalker 78 may include a crop stalker 78-1 and a crop stalker 78-2. In such a case, the crop stalker 78-1 is positioned on one of the lateral sides 80-1, 80-2, and the crop stalker 78-2 is positioned on the other of the lateral sides 80-1, 80-2.

The crop stalker 78-1 is a first type of crop stalker, and the crop stalker 78-2 is a second type of crop stalker. The first type of crop stalker 78-1 and the second type of crop stalker 78-2 are different from one another. For example, the first type of crop stalker 78-1 and the second type of crop stalker 78-2 may be oppositely sensed. The first and second types may be inversely related to one another such that the crop stalker 78-1 and the crop stalker 78-2 are configured in a laterally inverse relationship to one another.

The first type of crop stalker 78-1 may be a lateral peak, and the second type of crop stalker 78-2 may be a lateral valley. Each lateral peak may be configured, for example, as a laterally projecting bump, and each lateral valley may be configured, for example, as a laterally recessed dimple. The crop stalkers 78-1, 78-2 may be formed without lateral through-holes, although, it is to be appreciated that a laterally projecting bump or a laterally recessed dimple of the crop stalkers 78-1, 78-2 could have a small through-hole.

In the illustrated embodiment, each crop stalker 78 includes a crop stalker 78-1 in the form of a lateral peak configured, for example, as a laterally projecting bump on the first lateral side 80-1 or the second lateral side 80-2 and a corresponding inversely related crop stalker 78-2 in the form of a lateral valley configured, for example, as a laterally recessed dimple on the other of the first lateral side 80-1 or the second lateral side 80-2. In other embodiments, the body 70 may be perforated such that the crop stalkers 78 may be configured as holes extending laterally through the body 70.

Each array 76-1, 76-2 may include crop stalkers 78-1 of the first type and crop stalkers 78-2 of the second type, to enhance the roughness of the array 76-1, 76-2. The crop stalkers 78-1 and the crop stalkers 78-2 extend in laterally opposite directions such that the crop stalkers 78-1 project laterally into the respective bay 52 and the crop stalkers 78-2 are recessed laterally from the respective bay 52. Each array 76-1, 76-2 may thus include crop stalkers 78-1 in the form of lateral peaks configured, for example, as laterally projecting bumps and crop stalkers 78-2 in the form of valleys configured, for example, as laterally recessed dimples, with the laterally projecting bumps and the laterally recessed dimples extending in laterally opposite directions.

With respect to each array 76-1, 76-2, the rows of crop stalkers include alternating rows of oppositely sensed crop stalkers 78-1, 78-2, and the columns of crop stalkers comprise alternating columns of oppositely sensed crop stalkers 78-1, 78-2. The crop stalkers of the rows and the crop stalkers of the columns include crop stalkers 78-1 in the form of peaks configured, for example, as laterally-projecting bumps and crop stalkers 78-2 in the form of valleys configured, for example, as laterally recessed dimples. The row pattern alternates between a row of crop stalkers 78-1 and a row of crop stalkers 78-2. For example, with respect

to array **76-1**, row R1 has crop stokers **78-2**, row R2 has crop stokers **78-1**, row R3 has crop stokers **78-2**, and row R4 has crop stokers **78-1**. Similarly, the column pattern alternates between a column of crop stokers **78-1** and a column of crop stokers **78-2**. For example, with respect to array **76-1**, column C1 has crop stokers **78-2**, column C2 has crop stokers **78-1**, column C3 has crop stokers **78-2**, and column C4 has crop stokers **78-1**. Such pattern of alternating rows of oppositely sensed crop stokers and alternating columns of oppositely sensed crop stokers enhance the surface area roughness of the upper zone **88-1** of each array **76-1**, **76-2**.

The crop stokers of each diagonal D alternate between the crop stoker **78-1** and the crop stoker **78-2**. For example, diagonal D1 of array **76-1** includes in sequence stoker **78-2**, stoker **78-1**, stoker **78-2**, stoker **78-1**, and stoker **78-2**. Such an arrangement enhances the surface area roughness of the upper zone **88-1**.

The array **76** may be stamped into the body **70**. In such a case, the pattern of the array **76-1** and the pattern of the array **76-2** are laterally inversely related to one another, such that the crop stokers of the first array **76-1** and the crop stokers of the second array **76-2** are configured in a laterally inverse relationship to one another.

It is to be appreciated that the arrays **76-1**, **76-2** may be manufactured by any suitable method and not necessarily by a stamping operation (e.g., molding, casting, forming, forging, bending, machining, additive process, from two separate pieces of sheet metal or other parts joined together, to name but a few other methods). In such a case, the first array **76-1** and the second array **76-2** may be configured in a manner distinct from one another such that they are not necessarily inversely related, in which case their patterns may be the same or different. For example, in some embodiments, it may be that the crop divider **50** has a single array of crop stokers on only one lateral side and not the other lateral side. It is to be appreciated that one or both arrays **76-1**, **76-2** may have crop stokers of a single type or more than two types.

In some embodiments, the crop divider **50** may include an attachment (not shown) coupled to the panel **58**. The attachment may include one or both of the arrays **76-1**, **76-2**. The attachment may be configured, for example, as an over-the-top cover on the panel **58**. The attachment may be coupled to the panel **58** in a wide variety of ways (e.g., riveted, pinned, or bolted). The attachment may be made from a wide variety of materials (e.g., ultra-high molecular weight polyethylene).

An intermediate crop divider **150** may be mounted with the **30** for fore-aft reciprocating movement therewith. The crop divider **150** may be positioned, for example, between a pair of crop dividers **50**, or between a crop divider **50** and a side edge of the pan **30**. The crop divider **150** extends in a fore-aft manner and above the stepped floor **46** to partition crop material laterally.

The crop divider **150** includes a panel **158** and a pair of legs **159** mounted to the stepped floor **46** and supporting the panel **158** above the stepped floor **46**. The panel **158** includes a body **170** and a periphery **172** disposed about the body **170**. The body **170** includes a multi-dimensional, clustered array **176** of crop stokers **178** extending laterally to engage crop material to advance the crop material in the crop-processing direction **48** during fore-aft reciprocating movement of the pan **30**. The crop divider **150** is shorter in length than the crop dividers, and positioned closer to the rear of the pan **30** than the front.

The array **176** includes a first array **176-1** on a first lateral side **180-1** of the body **170** and a second array (not shown) on an opposite second lateral side (not shown) of the body **170**. The first and second arrays are inversely related to one another.

The crop stokers **178** are similar in structure and function as the crop stokers **78**. Each crop stoker **178** includes a crop stoker **178-1** and a crop stoker **178-2** laterally inverse to the crop stoker **178-1**. The crop stoker **178-1** is configured, for example, as a laterally projecting bump, and the crop stoker **178-2** is configured, for example, as a corresponding laterally recessed dimple. In other embodiments, the body **170** may be perforated such that the crop stokers **178** may be configured as holes extending laterally through the body **170**.

In the array **176** and each array **176-1**, **176-2**, the crop stokers **178** are arranged multi-dimensionally. For example, the crop stokers **178** are arranged in rows and columns. The rows alternate between rows of crop stokers **178-1** and rows of crop stokers **178-2**. Similarly, the columns alternate between columns of crop stokers **178-1** and columns of crop stokers **180-2**. As such, each diagonal sequentially alternates between crop stokers **178-1** and crop stokers **178-2**.

Rows and columns are staggered. For example, immediately adjacent rows are staggered relative to one another, and immediately adjacent columns are staggered relative to one another.

In some embodiments, the crop divider **150** may include an attachment (not shown) coupled to the panel **158**. The attachment may include one or both of the arrays **176-1**, **176-2**. The attachment may be configured, for example, as an over-the-top cover on the panel **158**. The attachment may be coupled to the panel **158** in a wide variety of ways (e.g., riveted, pinned, or bolted). The attachment may be made from a wide variety of materials (e.g., ultra-high molecular weight polyethylene).

In some embodiments, the crop divider **50** and/or the intermediate crop divider **150** may be applied to the return pan **40**.

While the above describes example embodiments of the present disclosure, these descriptions should not be viewed in a limiting sense. Rather, other variations and modifications can be made without departing from the scope and spirit of the present disclosure as defined in the appended claims.

What is claimed is:

1. A cleaning system for an agricultural combine harvester, the cleaning system comprising:

a pan comprising a stepped floor to receive crop material thereon, the pan arranged for fore-aft reciprocating movement to advance the crop material in a crop-processing direction, and

a crop divider mounted with the pan for fore-aft reciprocating movement therewith, the crop divider extending in a fore-aft manner and above the stepped floor to partition the crop material laterally, the crop divider comprising a body and a periphery disposed about the body, the body comprising a multi-dimensional, clustered array of crop stokers extending laterally to engage the crop material to advance the crop material in the crop-processing direction during fore-aft reciprocating movement of the pan.

2. The cleaning system of claim 1, wherein the array of crop stokers comprises crop stokers of a first type and crop stokers of a second type.

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3. The cleaning system of claim 2, wherein the first type is a lateral peak, and the second type is a lateral valley.

4. The cleaning system of claim 2, wherein the first type comprises laterally projecting bumps, and the second type comprises laterally recessed dimples.

5. The cleaning system of claim 1, wherein the array comprises rows of crop stokers and columns of crop stokers.

6. The cleaning system of claim 5, wherein the crop stokers of each row are positioned at regular intervals, and the crop stokers of each column are positioned at regular intervals.

7. The cleaning system of claim 5, wherein the rows of crop stokers comprise alternating rows of oppositely sensed crop stokers, and the columns of crop stokers comprise alternating columns of oppositely sensed crop stokers.

8. The cleaning system of claim 7, wherein the alternating rows are alternatingly staggered relative to one another, and the alternating columns are alternatingly staggered relative to one another.

9. The cleaning system of claim 5, wherein the rows are alternatingly staggered relative to one another, and the columns are alternatingly staggered relative to one another.

10. The cleaning system of claim 9, wherein the crop stokers of the rows and the crop stokers of the columns comprise laterally-projecting bumps and laterally recessed dimples.

11. The cleaning system of claim 1, wherein the body comprises a first lateral side and an opposite second lateral side, and each crop stoker provides a laterally projecting bump on the first lateral side or the second lateral side and an inversely related laterally recessed dimple on the other of the first lateral side or the second lateral side.

12. The cleaning system of claim 1, wherein the crop divider partitions the pan into a first bay and a second bay, the body comprises a first lateral side and an opposite second

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lateral side, the array is positioned on the first lateral side and extends laterally to engage crop material in the first bay, the body comprises a multi-dimensional, clustered second array of crop stokers positioned on the second lateral side and extends laterally to engage crop material in the second bay, and each of the array and the second array comprises rows and columns of laterally projecting bumps and laterally recessed dimples.

13. The cleaning system of claim 1, wherein the array of crop stokers comprises laterally projecting bumps.

14. The cleaning system of claim 1, wherein each crop stoker of the array of crop stokers has a maximum size, and immediately adjacent crop stokers are spaced apart from one another by an offset less than the maximum size.

15. The cleaning system of claim 1, wherein the crop divider has an X dimension parallel to the stepped floor and a Y dimension orthogonal to the stepped floor, the stepped floor has a step height in the Y dimension, the body of the crop divider comprises an upper zone with the array of crop stokers and a lower zone defined in the X dimension by a length of the array of crop stokers and in the Y dimension by at least a distance between the stepped floor and the array, and the distance is greater than or equal to a value of the step height.

16. The cleaning system of claim 1, wherein the crop divider comprises an upstream half and a downstream half downstream of the upstream half in the crop-processing direction, and the downstream half comprises the array of crop stokers.

17. The cleaning system of claim 1, wherein the crop divider has a Y dimension orthogonal to the stepped floor, and the array of crop stokers increases in height in the Y dimension from the stepped floor as the array extends in the crop-processing direction.

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